

Precision position sensing guide: Technology comparison

Application Note

Precision position sensing guide: Technology comparison

Application Note No. AN001092



Valid for:
All position sensors from ams OSRAM

Abstract

This application note provides a comprehensive guide for selecting the optimal position sensing technology—inductive or magnetic (Hall-based)—from ams OSRAM portfolio. It outlines the core differences in sensing principles, design integration, performance characteristics, and application suitability. The document includes a detailed comparison table highlighting key parameters such as target type, sensing distance, stray field immunity, packaging, and functional safety. Specific use cases in automotive and industrial domains are evaluated, illustrating the decision-making process for sensor selection based on environmental, mechanical, and performance criteria. The guide serves as a practical reference for engineers and system designers seeking high-precision, robust, and space-efficient position sensing solutions across a wide range of applications.

Table of contents

- 1 Precision position sensing guide 4
 - 1.1 Selecting between inductive and magnetic sensor technologies 4
 - 1.2 Quick technology comparison 6
- 2 Revision information 9

1 Precision position sensing guide

1.1 Selecting between inductive and magnetic sensor technologies

This application note helps you choose the right position sensor technology from ams OSRAM, by comparing inductive and magnetic (Hall-based) sensors based on application fit, performance, and integration requirements.

Figure 1: Magnetic position sensor

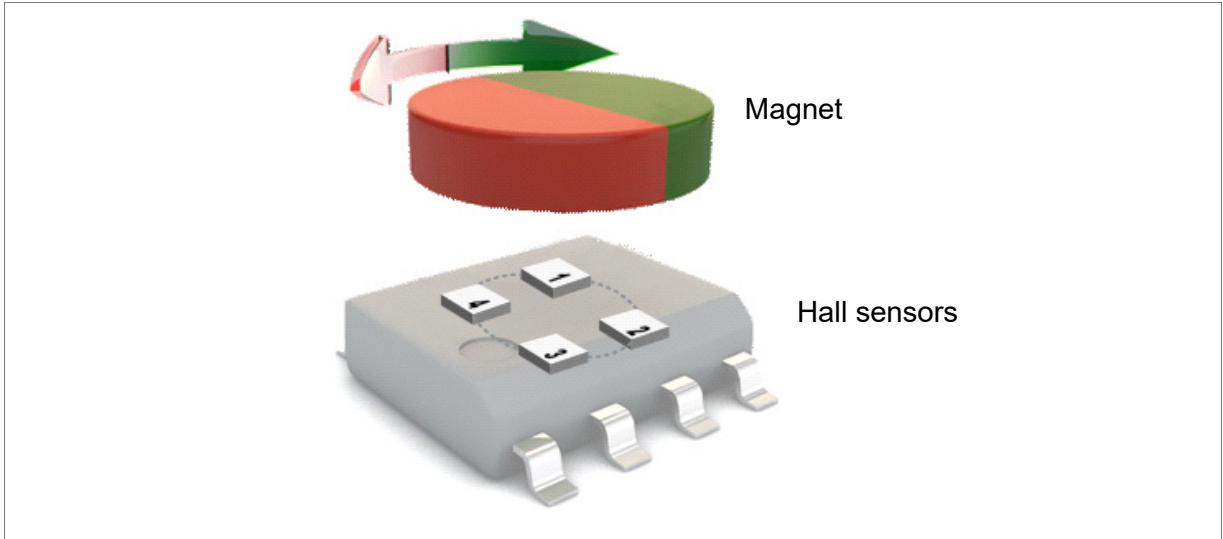


Table 1: Description of magnetic position sensor

Parameter	Description
Sensing element	Integrated hall sensors
Target	Small permanent magnet (Ø6–8 mm, thickness ~2.5 mm)
Operation	Magnet is attached to the rotating or moving part; signal is processed by an analog frontend and optionally a digital backend to provide angular output.

Figure 2: Inductive position sensor

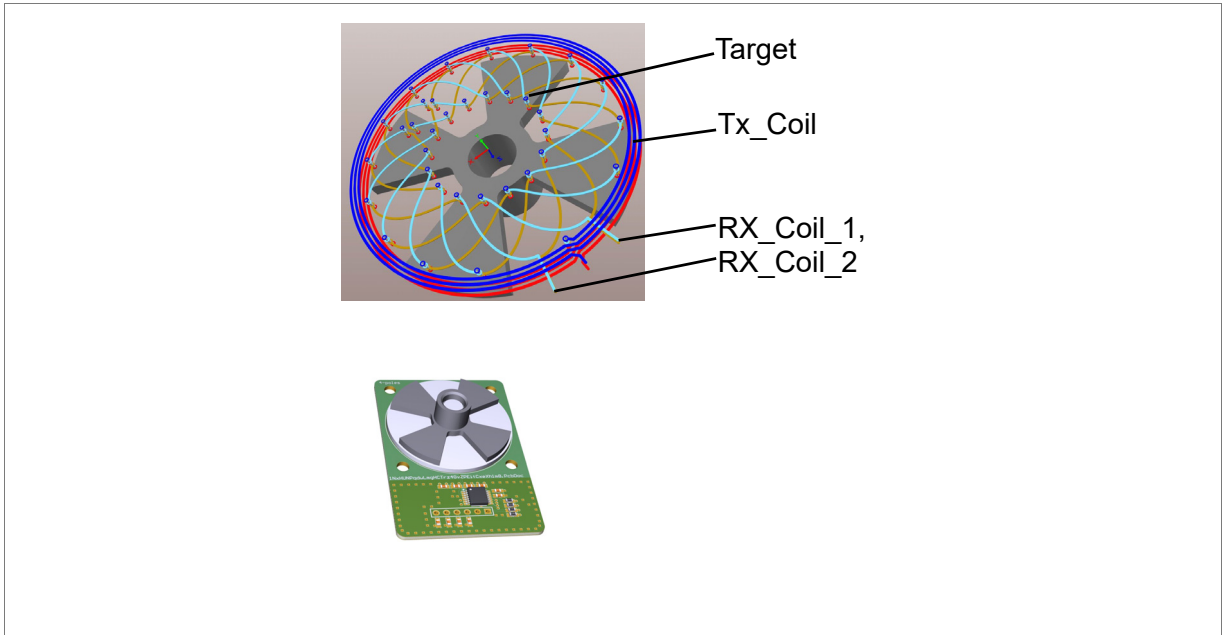
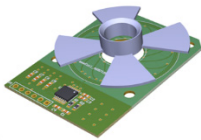
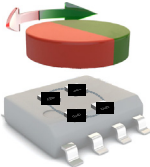


Table 2: Description of inductive position sensor

Parameter	Description
Sensing element	RX_Coil_1 and RX_Coil_2
Target	Conductive material (e.g., steel, aluminum, copper)
Operation	TX_Coil is driven by a ~4 MHz signal; voltage is induced in the RX coils based on the position of the target

1.2 Quick technology comparison

Table 3: Comparison of inductive and magnetic sensors

Feature	Inductive sensor	Magnetic sensor
		
Target type	Conductive metal	Permanent magnet
Target material	Metallic only	Any (with magnet)
Contactless	Yes	Yes
Integration	Custom coil with flexible sized PCB	Compact standard ICs
Stray field immunity	Excellent	Excellent
Sensing distance	Short range	Longer, non-contact
Packaging	TSSOP14	TSSOP14, SOIC8, MLF-16/32/40, SIP, TQFP-2, QFN-16, WLCSP
Functional safety/ASIL	ASIL-C (ASIL-D possible with redundant coil design and sensor)	Up to ASIL-D
Motion	On-axis, Off-axis, Linear	On-axis
Accuracy (typical)	$\pm 0.075^\circ$ (4 pole pairs and 90°) $\pm 0.0375^\circ$ (8 pole pairs and 90°)	$\pm 0.5^\circ$ (1 pole pair and 360°)
Size (typical)	30 x 15 mm (incl. PCB and passive components) smaller is possible but accuracy will suffer	1.5 x 10 mm (incl. PCB and passive components)
Output interface	Analog (sin/cos)	Analog (sin/cos, linear), SENT, I ² C, SPI, ABI, UVW, PWM, PSI5

1.2.1 When to choose inductive sensors

Inductive sensors are optimal when:

- High-precision detection of metallic targets is needed
- Environmental noise or stray magnetic fields are present
- You need robust, short-range sensing in harsh conditions
- Space is limited along the length of motor shaft or housing (low profile applications where sensor needs to fit in a thin, compressed space)
- Looking for easy replacement of AMR and GMR sensors

1.2.2 When to choose magnetic sensors

Magnetic sensors are preferred when:

- Sensing is needed through plastic, aluminum, or non-metal materials
- Space and integration constraints favor compact solutions
- Multiple programmable interface options are needed
- Environmental noise or stray magnetic fields are present

1.2.3 Automotive use case highlights

Table 4: Automotive use case highlights

Application	Recommended sensor type	Why
Electric power steering (EPS)	Inductive or magnetic	Precision, space constraint
Hydraulic pump motor	Magnetic	High speed, sealed design
Chassis position sensing (CPS)	Magnetic	Through cover detection, compact form
Traction motors	Inductive	Harsh environment, precision, compact form factor

1.2.4 Industrial use case highlights

Table 5: Industrial use case highlights

Application	Recommended sensor type	Why
Factory automation	Inductive	Metal object detection, immunity to noise
Conveyor position feedback	Magnetic	Non-contact sensing, easy installation
Robotic joint monitoring	Inductive	High Precision

1.2.5 Decision matrix summary

Table 6: Decision matrix summary

Scenario	Best choice
High-speed motors	Both
Metal detection in harsh environments	Inductive
Plastic or non-metal detection	Magnetic
Plastic or non-metal housing	Both
Safety critical (ASIL-C/D) redundancy	Both
Resolver replacement	Inductive

2 Revision information

Changes to current revision v1-00	Page
Initial production version	
• Page and figure numbers for the previous version may differ from page and figure numbers in the current revision. • Correction of typographical errors is not explicitly mentioned.	

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